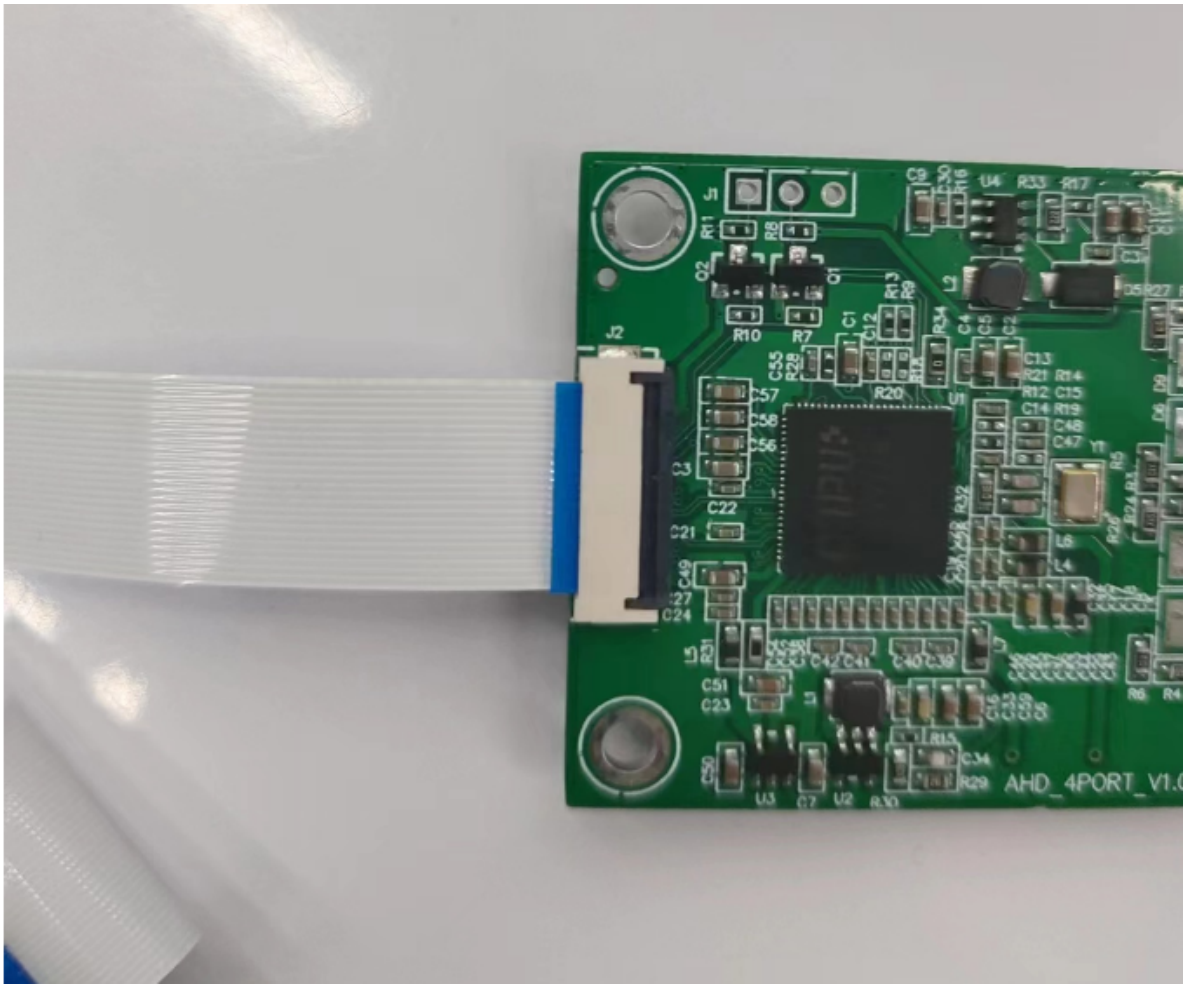
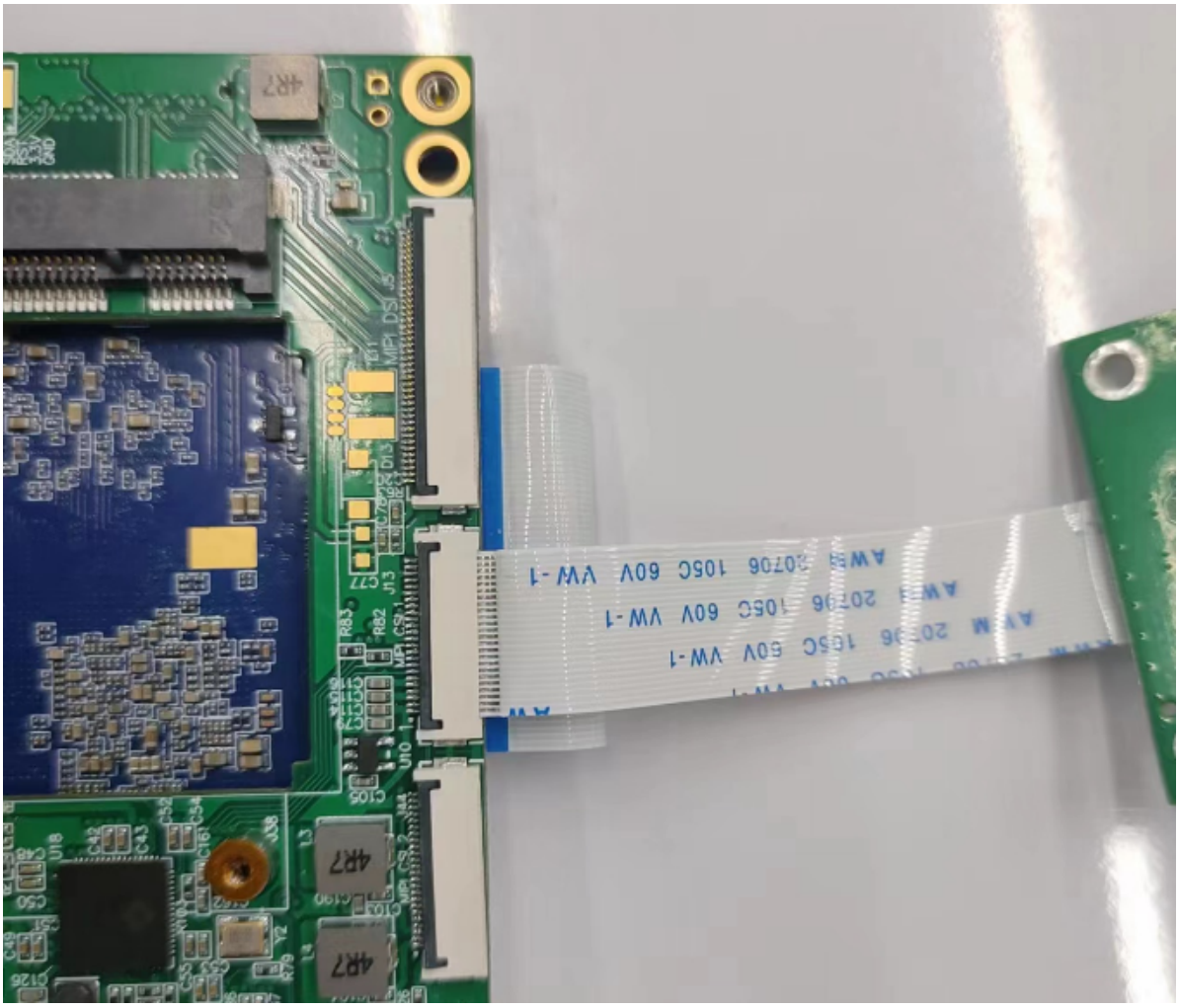
1. AHD小板与主板的连接

主板通过CSI接口连接CSI转AHD小板，可以支持同时接入4路AHD摄像头（RK3588 XB-NANO主板有两路CSI可以接入8路AHD摄像头）。最大支持1920x1080/25FPS。

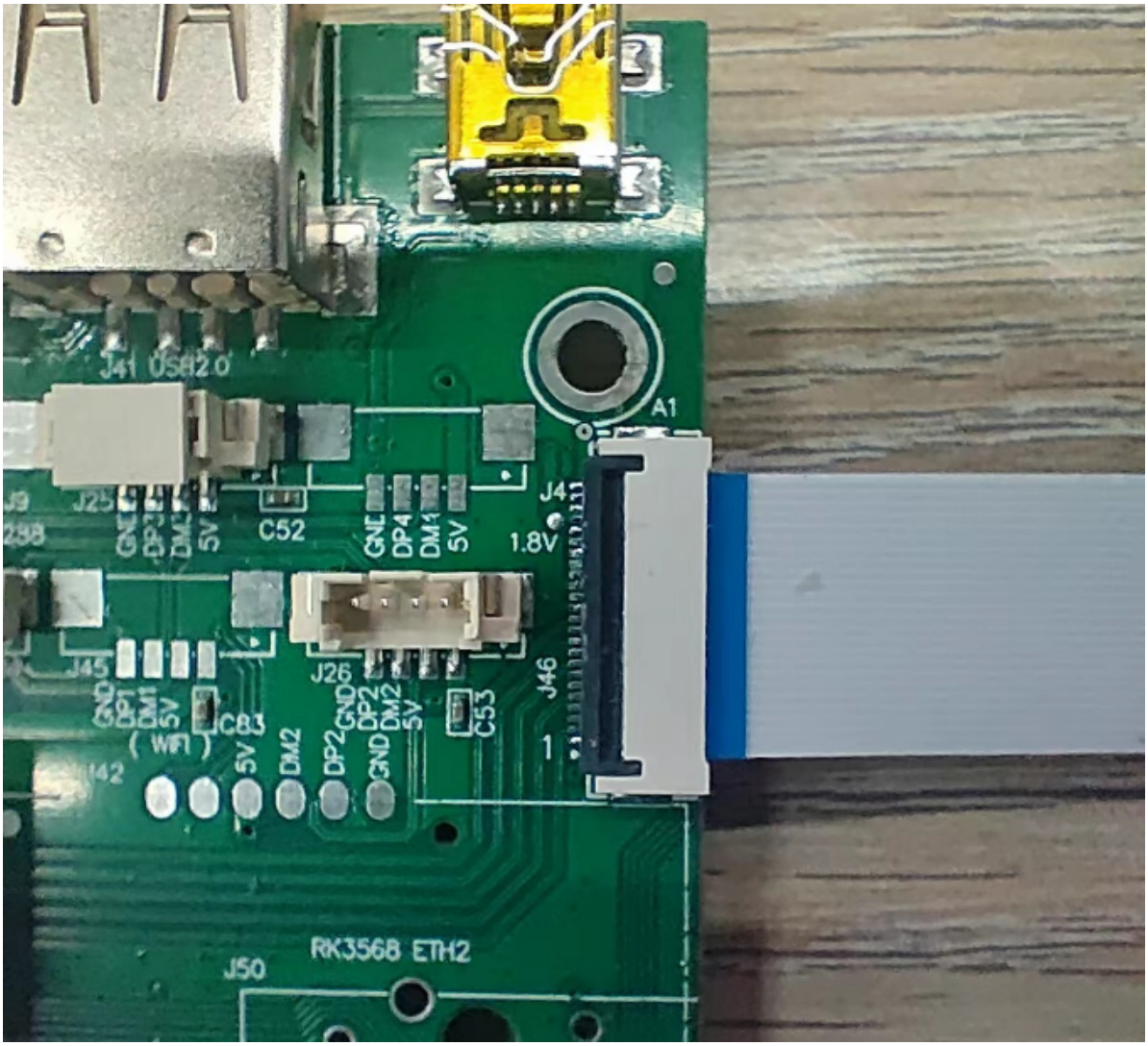
1. AHD小板端



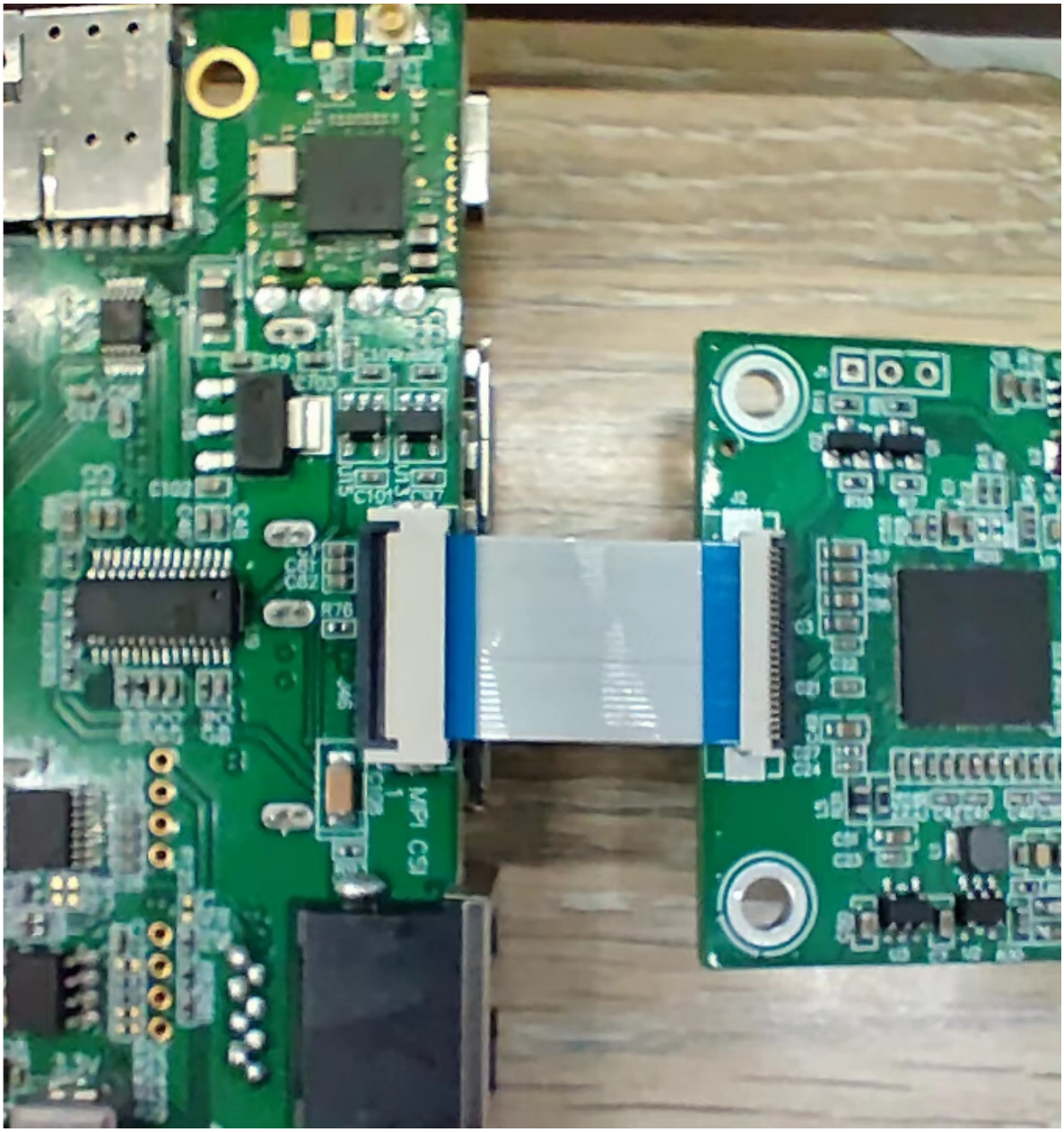
2. XB-NANO主板的连接



3. XB主板的连接



4. XB-PLUS主板的连接



2. 相机分辨率配置

AHD 小板支持1920x1080 和 1280x720 两种分辨的相机，需要修改配置文件支持对应分辨率的相机，不支持同时使用两种不同分辨率的相机。

修改配置文件 `/etc/mstahd4port.conf`，如果这个文件不存在就创建一个

- 1920x1080　编辑内容为 `camode=1920x1080-25`
- 1280x720　 编辑内容为 `camode=1280x720-25`

3. v4l2-ctl 命令使用

1. 查看相机节点

`v4l2-ctl --list-devices`

```
root@rk3588:/# v4l2-ctl --list-devices
rkcif-mipi-lvds2 (platform:rkcif):
        /dev/media0
        /dev/media1

rkcif (platform:rkcif-mipi-lvds2):
        /dev/video0
        /dev/video1
        /dev/video2
        /dev/video3
        /dev/video4
        /dev/video5
        /dev/video6
        /dev/video7

rkcif (platform:rkcif-mipi-lvds4):
        /dev/video8
        /dev/video9
        /dev/video10
        /dev/video11
        /dev/video12
        /dev/video13
        /dev/video14
        /dev/video15
```

如上图所示:

CSI1对应的4个相机节点分别是`/dev/video0` `/dev/video1` `/dev/video2` `/dev/video3`

CSI1对应的4个相机节点分别是`/dev/video8` `/dev/video9` `/dev/video10` `/dev/video11`

2. 查看相机支持的分辨率

`v4l2-ctl --list-formats-ext --device /dev/video0`

```
root@rk3588:/# v4l2-ctl --list-formats-ext --device /dev/video0
ioctl: VIDIOC_ENUM_FMT
        Type: Video Capture Multiplanar

        [0]: 'NV16' (Y/CbCr 4:2:2)
                Size: Stepwise 64x64 - 1920x1080 with step 8/8
        [1]: 'NV61' (Y/CrCb 4:2:2)
                Size: Stepwise 64x64 - 1920x1080 with step 8/8
        [2]: 'NV12' (Y/CbCr 4:2:0)
                Size: Stepwise 64x64 - 1920x1080 with step 8/8
```

3. v4l2-ctl 取流, ffplay播放

v4l2-ctl 取流

```
v4l2-ctl --verbose -d /dev/video0 --set-fmt-
video=width=1920,height=1080,pixelformat=NV12 --stream-mmap=3 --stream-
to=mstahd.yuv --stream-count=100 --stream-poll
```

ffplay播放

```
ffplay -f rawvideo -video_size 1920x1080 -pix_fmt nv12 mstahd.yuv
```

4. gst-launch 命令使用

1. 预览

```
export DISPLAY=:0
export XAUTHORITY=/home/marvsmart/.Xauthority
export LD_LIBRARY_PATH=$LD_LIBRARY_PATH:/usr/lib/aarch64-linux-gnu/gstreamer-1.0
SINK=ximagesink

gst-launch-1.0 v4l2src device=/dev/video0 ! video/x-
raw,format=NV12,width=1920,height=1080, framerate=20/1 ! videoconvert ! $SINK &

wait
```

2. 取流后编码为H264保存

```
gst-launch-1.0 v4l2src device=/dev/video0 num-buffers=500 ! videoconvert !
video/x-raw,format=NV12,width=1920,height=1080 ! queue ! mpph264enc ! queue !
h264parse ! qtmux ! filesink location=h264.mp4
```

3. 取流后编码推流

```
gst-launch-1.0 v4l2src device=/dev/video0 ! videoconvert ! video/x-
raw,format=NV12,width=1920,height=1080 ! queue ! mpph264enc ! queue ! h264parse !
flvmux streamable=true ! queue ! rtmpsink sync=false
location="rtmp://192.168.6.125"
```